Program 8

Create Kubernetes Pods using YAML Descriptors

Program Stucture:

program8/
-------->nodejs-configmap.yaml
-------->nodejs-pod.yaml
-------->server.js

Procedure:

step 1: Start Minikube
minikube start
Check the status of minikube
minikube status

```
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx: ~/program8
bash: /home/pavithra/dev/flutter/bin/cache: No such file or directory
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~$ mkdir program8
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~$ cd program8
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program8$ nano sevrver.js
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program8$ nano sevrver.js
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program8$ nano nodejs-configapp.yaml
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program8$ nano nodejs-pad.yaml
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program8$ nano nodejs-pod.yaml
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program8$ minikude start
minikude: command not found
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program8$ minikude start
minikude: command not found
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program8$ minikube start
   minikube v1.37.0 on Ubuntu 22.04
   Using the docker driver based on existing profile
   Starting "minikube" primary control-plane node in "minikube" cluster
   Pulling base image v0.0.48 ...
   Updating the running docker "minikube" container ...
   Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
   Verifying Kubernetes components...
    ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
   Enabled addons: storage-provisioner, default-storageclass

!  /usr/bin/kubectl is version 1.30.14, which may have incompatibilities with Kubernetes 1.34.0.
    ■ Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
   Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

step 2: Craete nodejs-configmap.yaml file and with the following code
nano nodejs-configmap.yaml

```
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx: ~/program8
  GNU nano 6.2                                          nodejs-configapp.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-app-config
data:
  server.js: |
    const http = require("http");
    const port = 3000;

    const server = http.createServer((req, res) => {
      res.end("Hello from Node.js running inside Kubernetes!");
    });

    server.listen(port, () => {
      console.log(`Server running at port ${port}`);
    });
```

step 3: Create nodejs-pod.yaml file with the following code
nano nodejs-pod.yaml

```
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx: ~/program8
  GNU nano 6.2                                          nodejs-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: program-8
  labels:
    app: nodejs
spec:
  containers:
    - name: nodejs
      image: node:18
      command: ["node", "/usr/src/app/server.js"]
      ports:
        - containerPort: 3000
      volumeMounts:
        - name: app-code
          mountPath: /usr/src/app
  volumes:
    - name: app-code
      configMap:
        name: nodejs-app-config
```

Step 4: create the server.js file with following code
nano server.js

```
const http = require("http");
const port = 3000;

const server = http.createServer((req, res) => {
  res.end("Hello from Node.js running inside Kubernetes!");
});

server.listen(port, () => {
  console.log(`Server running at port ${port}`);
});
```

step 5: Apply the yaml file

docker apply -f nodejs configmap.yaml

get the pods

docker get pods

step 6 : Access the nodejs application

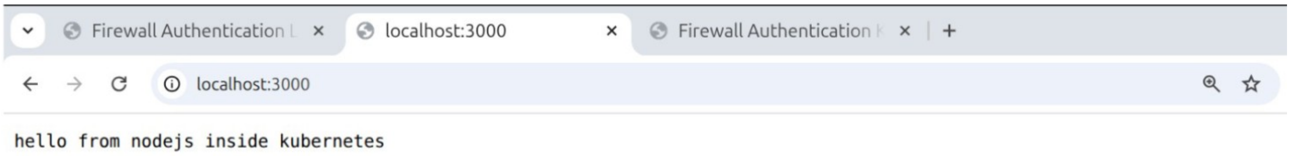
kubectl port-forward pod/program8 3000:3000

```
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program8$ nano nodejs-configapp.yaml
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program8$ kubectl apply -f nodejs-configapp.yaml
configmap/nodejs-app-config created
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program8$ kubectl apply -f nodejs-pod.yaml
pod/program-8 created
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program8$ kubectl get pods
NAME        READY   STATUS              RESTARTS   AGE
program-8   0/1     ContainerCreating   0          22s
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program8$ kubectl get pods
NAME        READY   STATUS    RESTARTS   AGE
program-8   1/1     Running   0          5m14s
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program8$ kubectl port-forward pod/program-8 300
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000
Handling connection for 3000
```

step 7: Acess on the browser

http://localhost:300

Out put


Firewall Authentication
localhost:3000
Firewall Authentication
localhost:3000
hello from nodejs inside kubernetes